

C3

Project Site Conditions

The project site sits atop the platform deck spanning active Long Island Rail Road tracks below. Foundation loads must transfer through the platform structure to caissons bearing on Manhattan schist bedrock at varying depths. Column locations may be constrained by the geometry of the rail infrastructure below — the diagrid structural system addresses this constraint by distributing loads across multiple bearing points rather than concentrating them on a conventional column grid. The site is essentially flat, elevated above the natural street grade of surrounding blocks. Utilities are routed through the platform from surrounding street infrastructure. Site access for construction and ongoing loading is constrained by limited vehicular entry points into the development. Wind conditions at grade are intensified by the venturi effect between adjacent supertall towers, directly affecting the building's entry sequence and base conditions.

Key Considerations

- Platform deck over active LIRR tracks: structural load transfer to bedrock caissons
- Column locations constrained by rail geometry; diagrid distributes loads across multiple points
- Flat site elevated above natural surrounding street grades
- Utilities routed through platform from surrounding street infrastructure
- Intensified ground-level wind due to venturi effect between supertall towers

